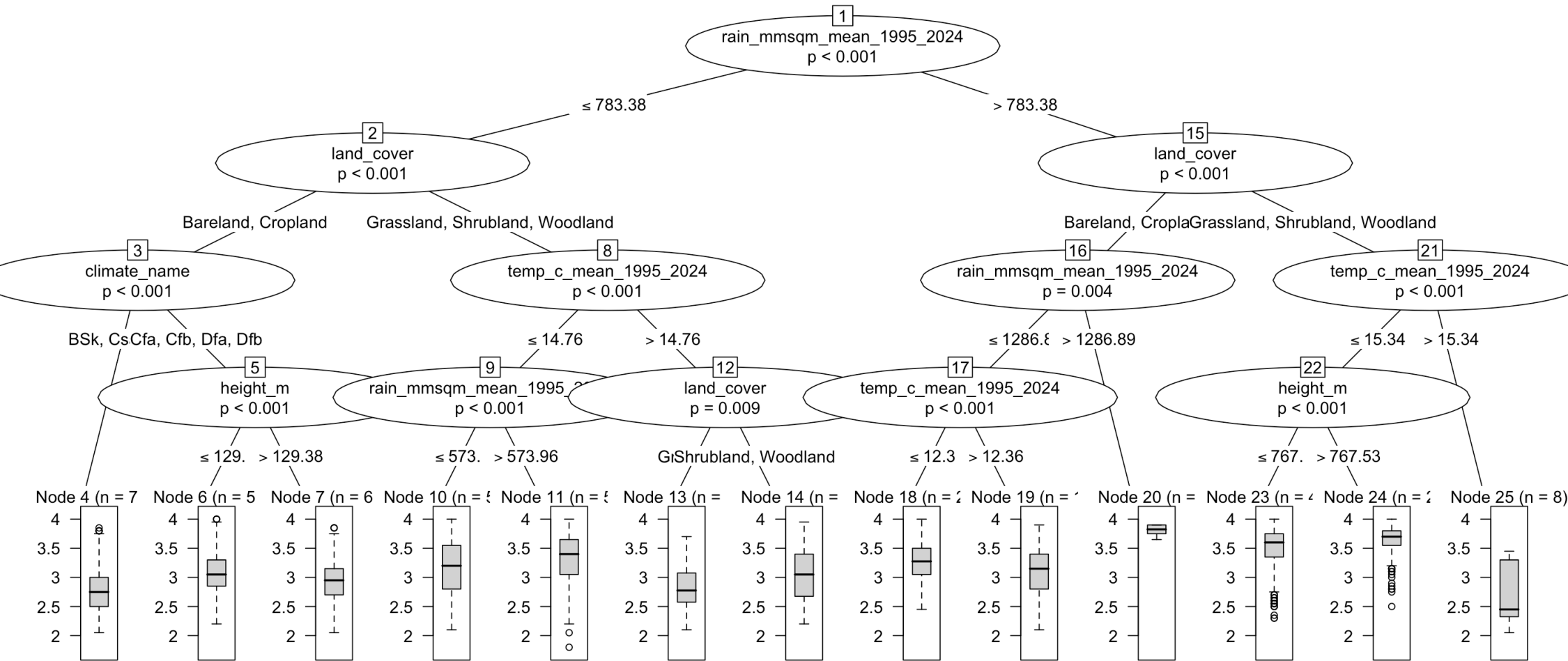


Conditional Inference Tree für SHI (maxdepth=4)



Relative Wichtigkeit der Einflussfaktoren auf den SHI

Conditional Inference Forest — native kategorische Verarbeitung

Einflussfaktor

rain_mmsqm_mean_1995_2024

land_cover

temp_c_mean_1995_2024

climate_name

height_m

land_use

0

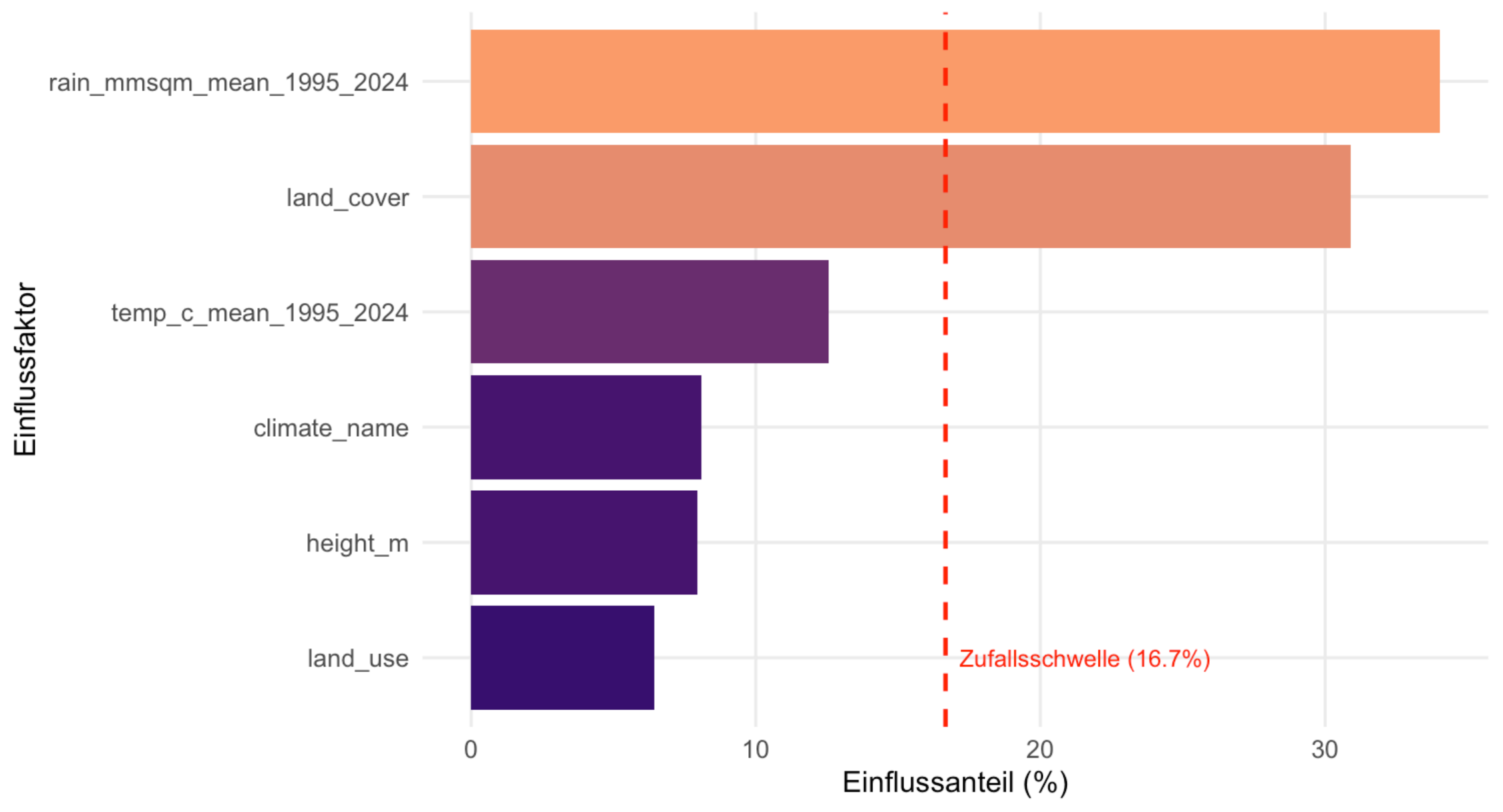
10

20

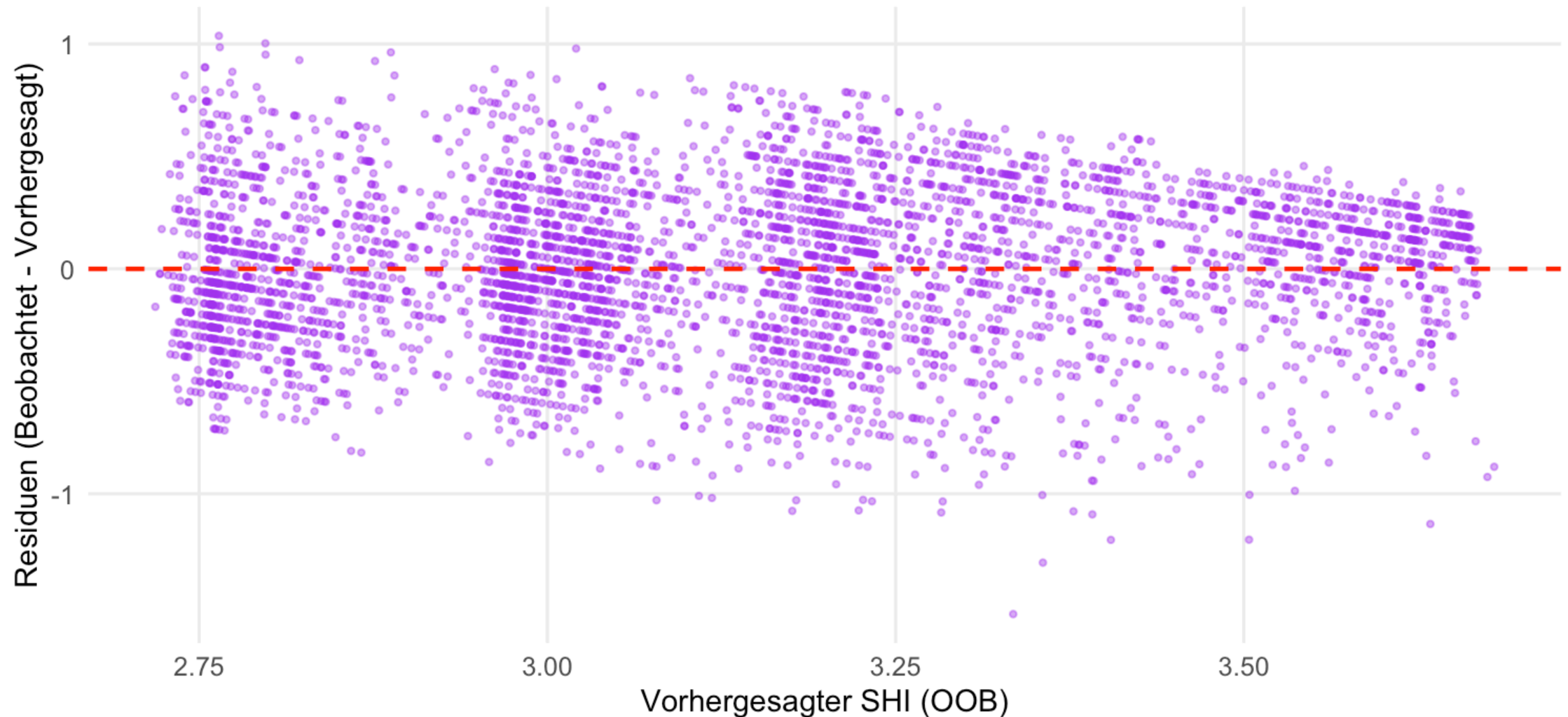
30

Einflussanteil (%)

Zufallsschwelle (16.7%)



Residuenanalyse des Conditional Inference Forest



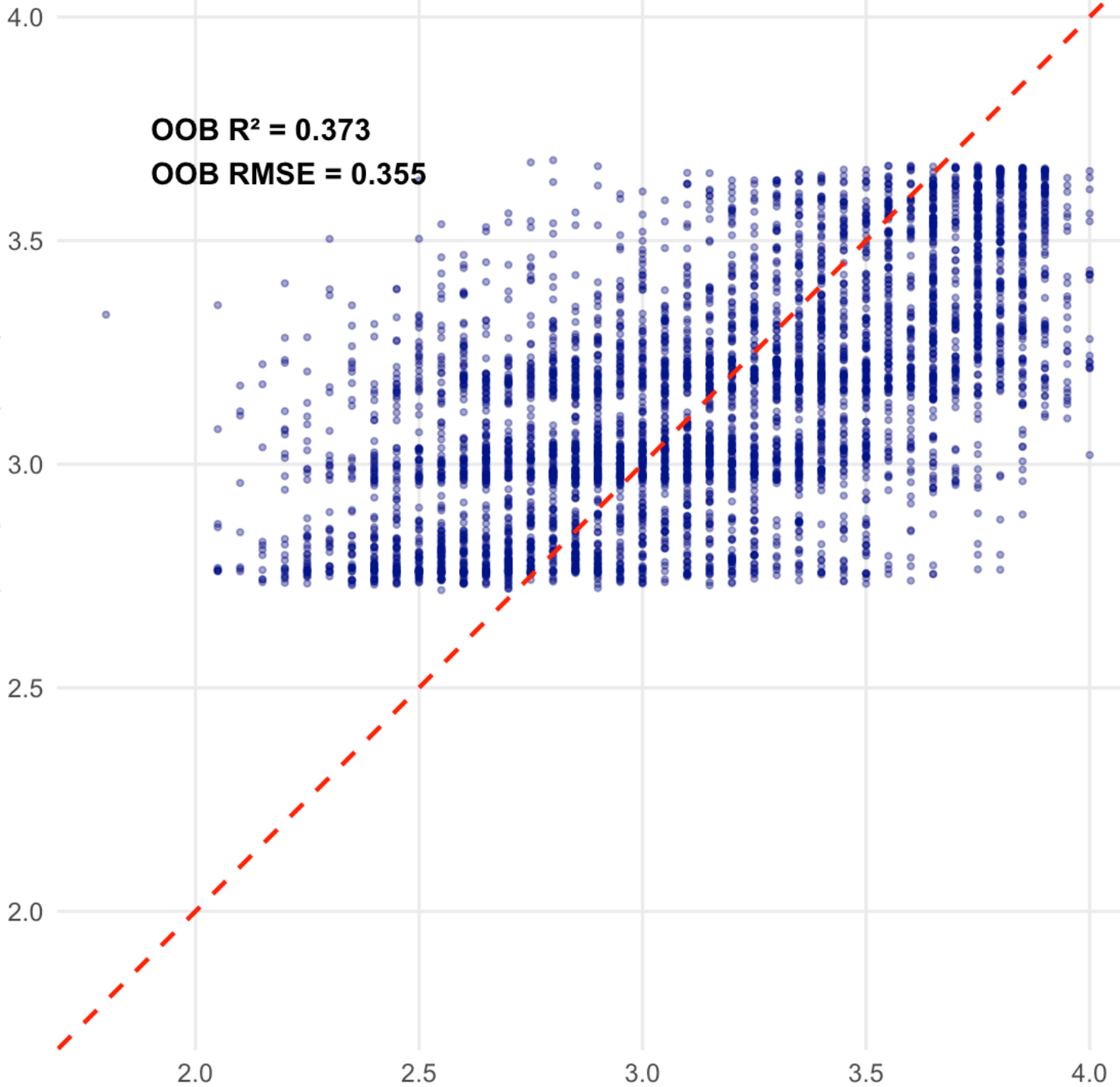
OOB-Vorhersagen vs. beobachteter SHI

OOB $R^2 = 0.373$

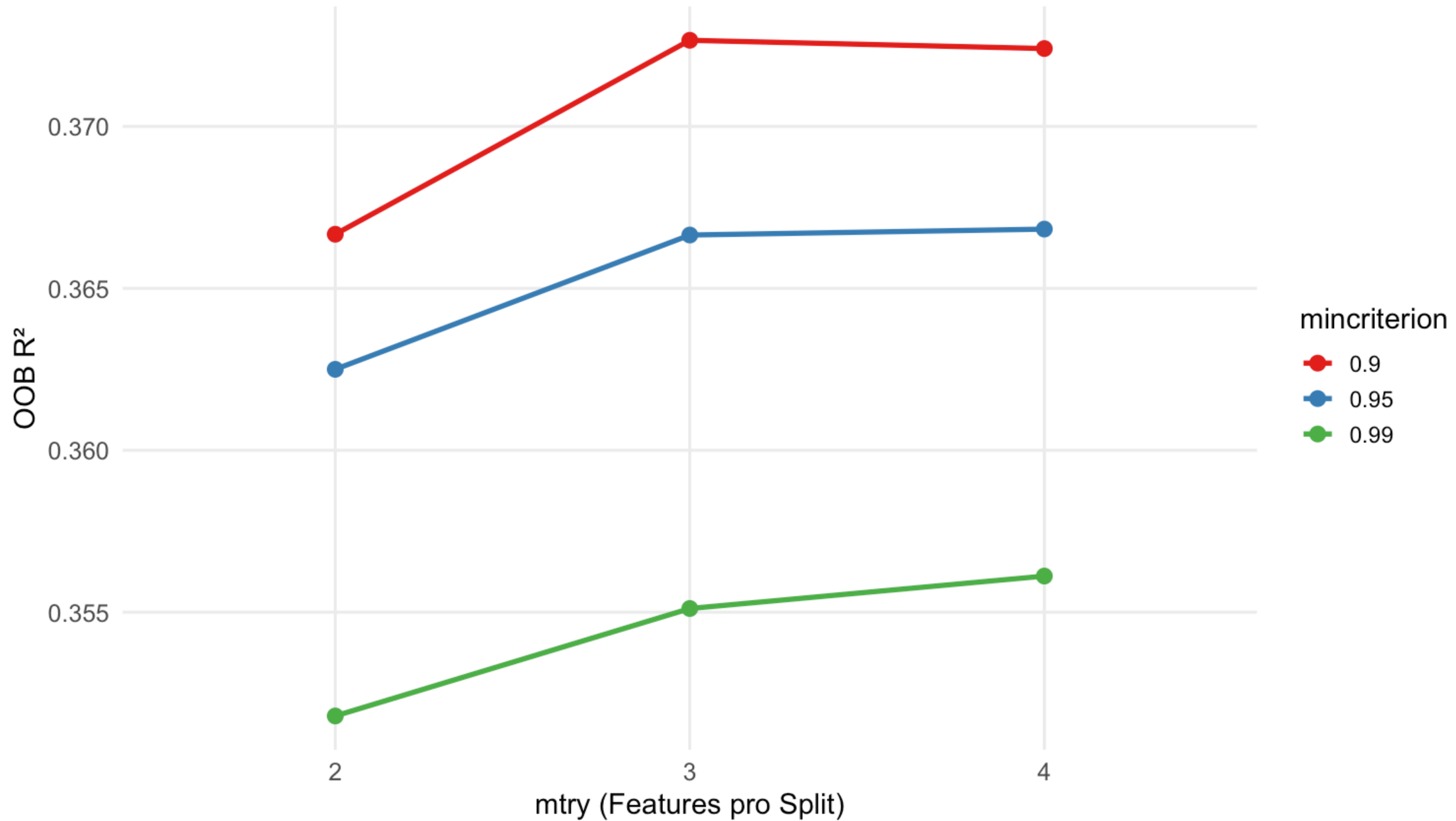
OOB RMSE = 0.355

Vorhergesagter SHI (OOB)

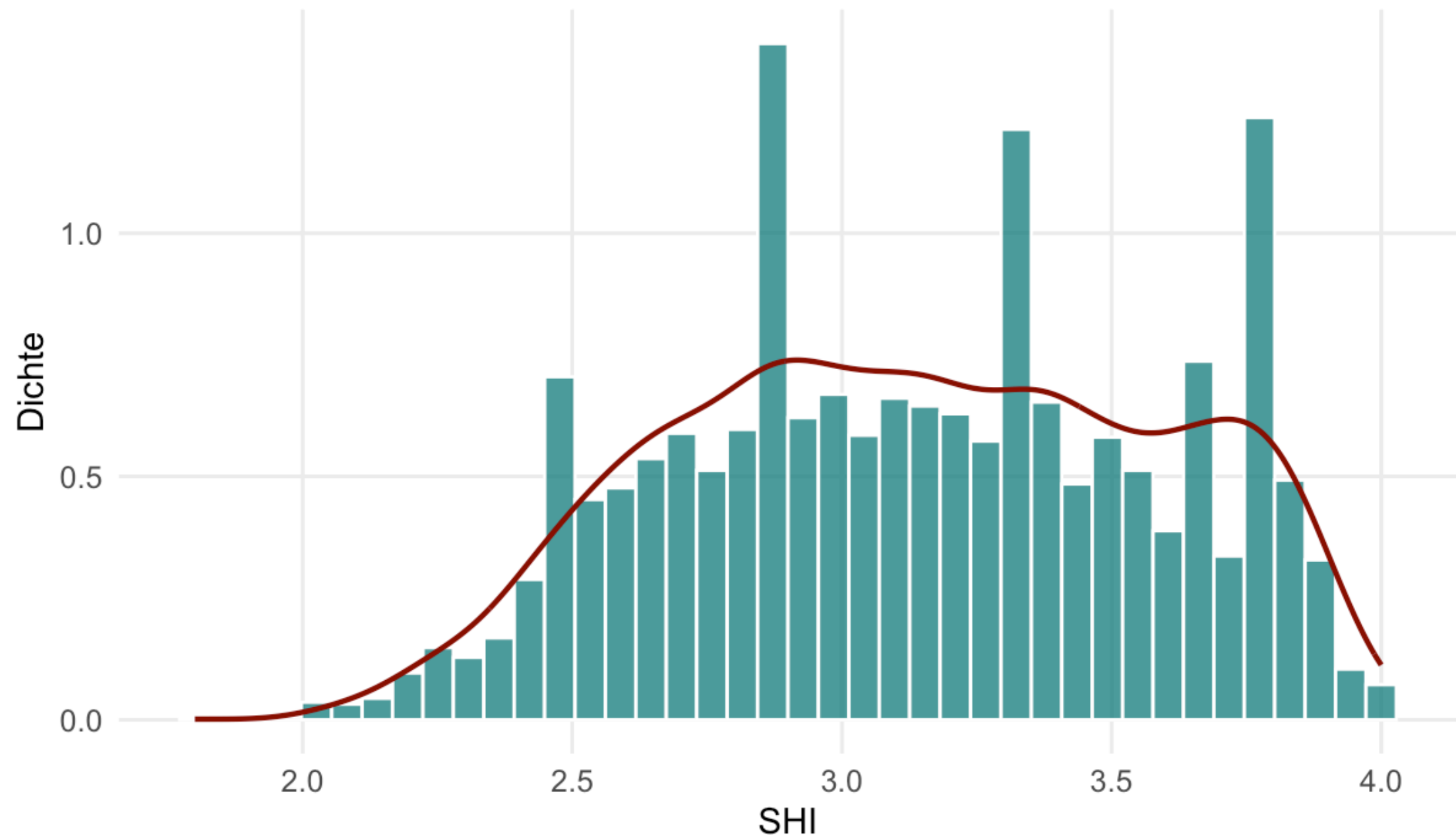
Beobachteter SHI



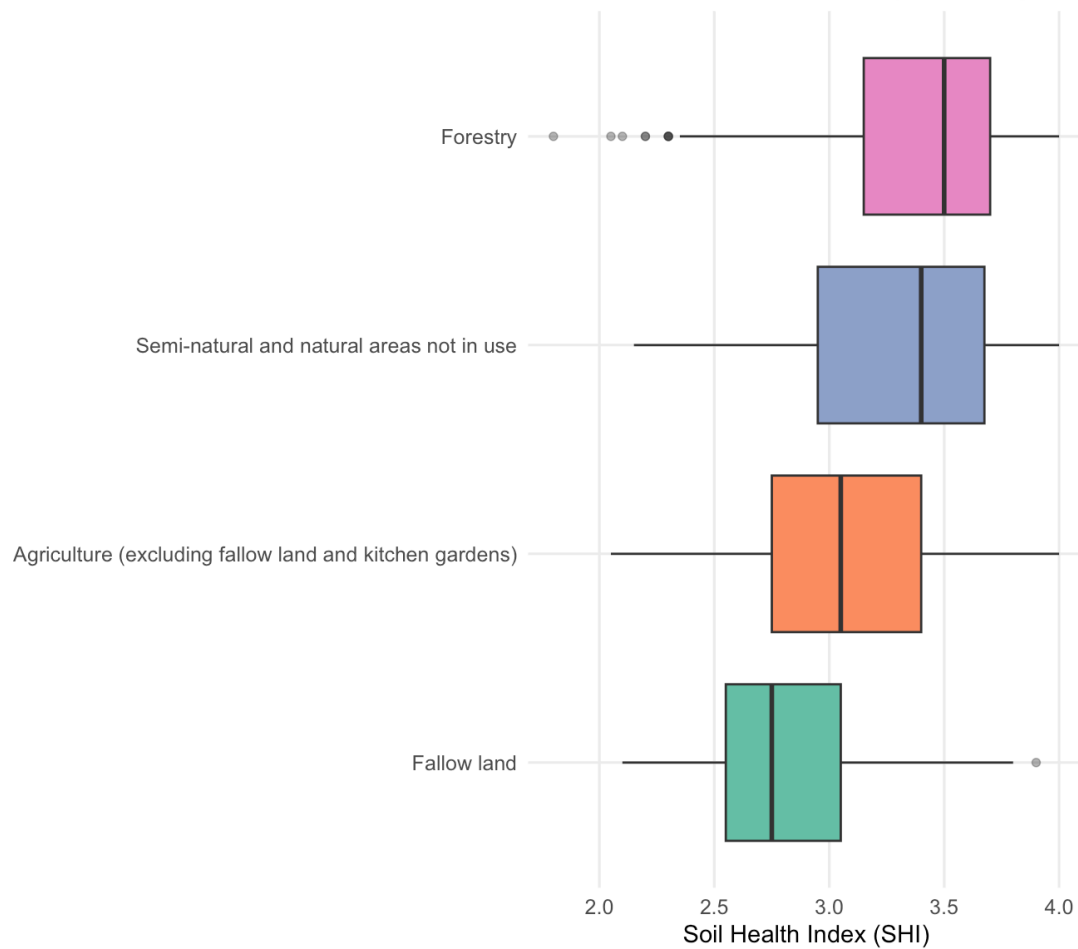
Modellgüte (OOB R^2) nach Hyperparametern



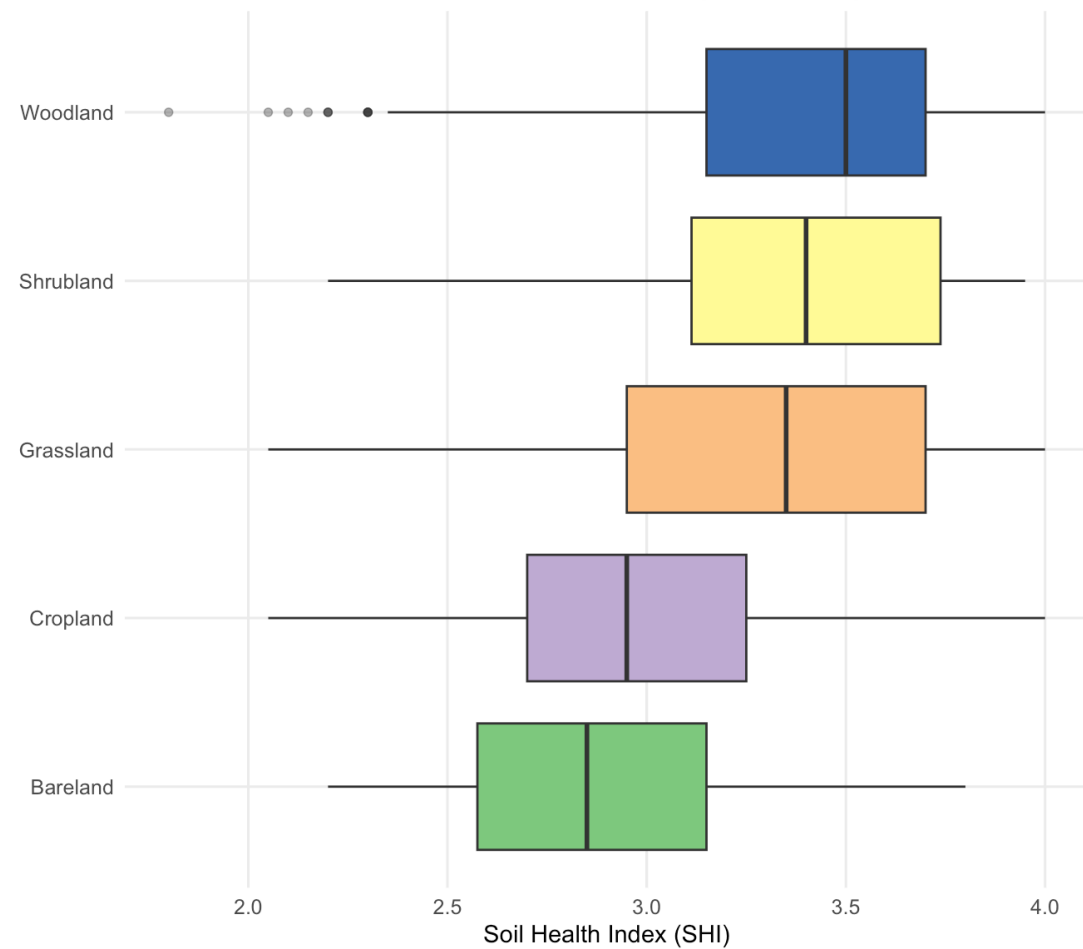
Verteilung des Soil Health Index (SHI)



SHI nach Landnutzung (land_use)



SHI nach Landbedeckung (land_cover)



Bodengesundheit (SHI) nach Köppen-Geiger-Klimaklasse

Klimaklasse

Cfb Temperate, no dry season, warm summer

Dfb Cold, no dry season, warm summer

Csb Temperate, dry summer, warm summer

Dfa Cold, no dry season, hot summer

Cfa Temperate, no dry season, hot summer

Csa Temperate, dry summer, hot summer

BSk Arid, steppe, cold

2.0

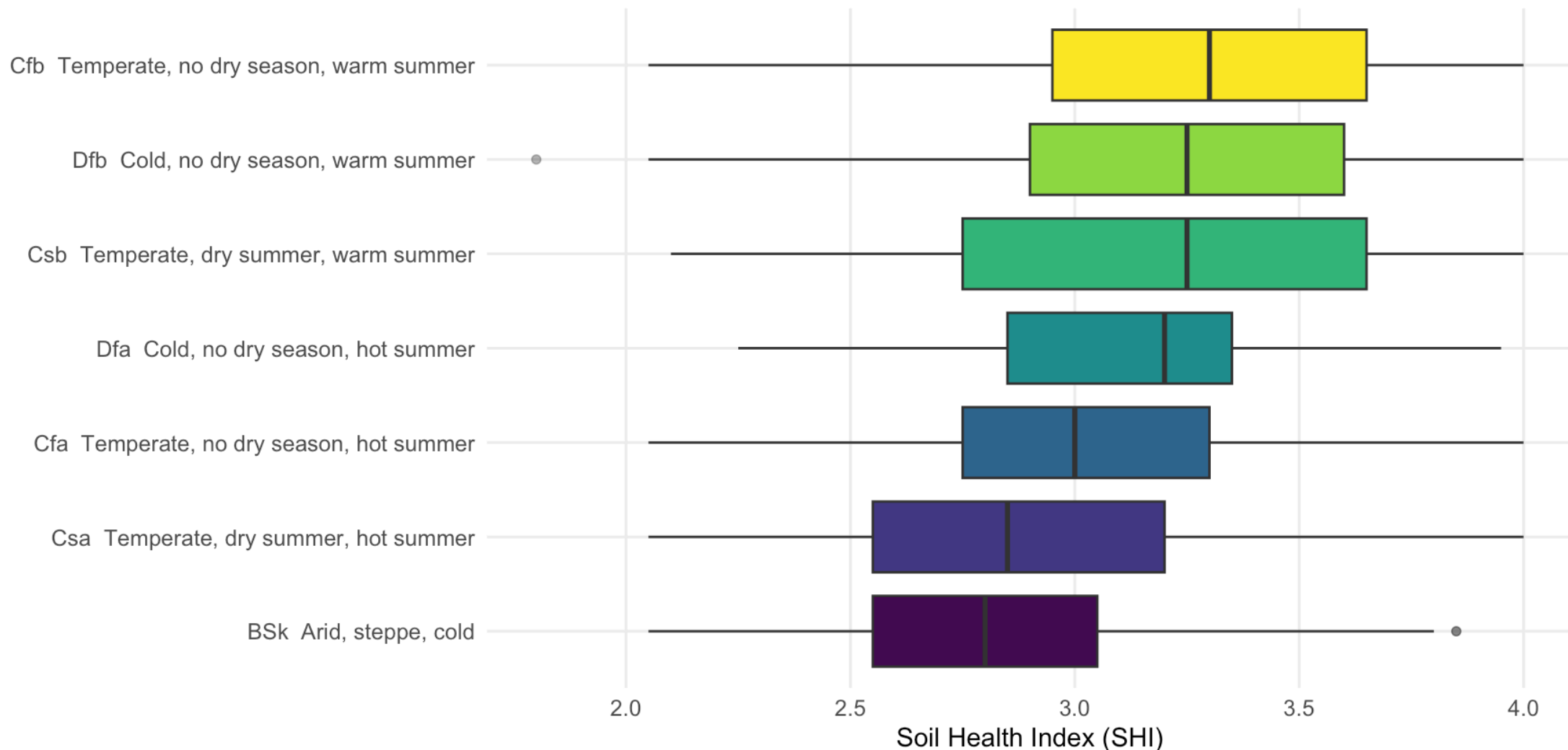
2.5

3.0

3.5

4.0

Soil Health Index (SHI)



Korrelationsmatrix der numerischen Einflussfaktoren

